

- 1 (a) Some students decided to find the respiratory quotient (RQ) of germinating pea seeds.

Pea seeds germinate best at temperatures between 13 °C to 18 °C.

Table 1.1 shows the main biological molecules in pea seeds that could be used to release energy for seed germination.

Table 1.1

biological molecule	g per 100 g of dry mass
carbohydrate	68
protein	25
fat	1

- (i) Predict the RQ value for these seeds in aerobic conditions.

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The students used the respirometer shown in Fig. 1.1.

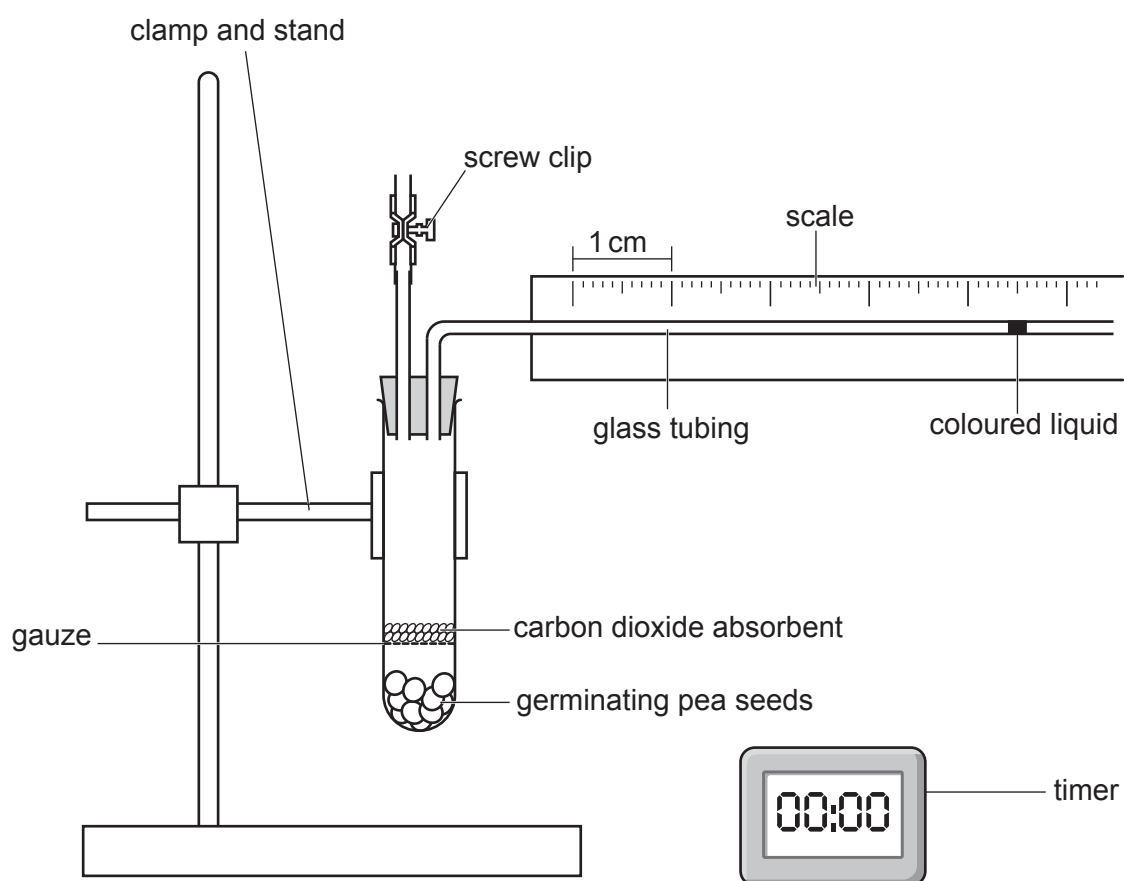


Fig. 1.1

- (b) Describe how the students should process their results to find the volumes of gases needed to calculate the RQ of germinating pea seeds.

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- (c) (i) The students predicted that peas could also respire in anaerobic conditions, producing alcohol and carbon dioxide.

Outline how the students could use the respirometer in Fig. 1.1 in anaerobic conditions to find out if their prediction was correct.

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..... [3]

- (ii) Table 1.2 shows the distance moved by the coloured liquid in anaerobic conditions in five different trials.

Table 1.2

trial	1	2	3	4	5
distance moved / mm	25	38	21	27	23

The students calculated a mean from the results in Table 1.2. They did not include one of the results in their calculation.

Identify the result that the students did not use **and** explain why they did not use it.

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[Total: 15]